Room Number _____

Seat Number _____

Student Number |__|__|__|__|__|__|__|__|__|

ANONYMOUSLY MARKED

(Please do not write your name on this exam paper)

THE UNIVERSITY OF SYDNEY

CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Mathematics and Statistics

EXAMINATION

Semester 1 - Main, 2018

MATH2921 Vector Calculus and Differential Eqs (Adv)

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:
This is a RESTRICTED OPEN book examination - specified materials permitted

During reading time - writing is not permitted at all

MATERIALS PERMITTED IN THE EXAM VENUE:
(No electronic aids are permitted e.g. laptops, phones)

One A4 sheet of handwritten and/or typed notes double sided

MATERIALS TO BE SUPPLIED TO STUDENTS:
2 x 16-page answer book

INSTRUCTIONS TO STUDENTS:

Please tick the box to confirm that your examination paper is complete. ☐

For Examiner Use Only

Q	Mark
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Total _____

1. Find all real critical points of the function

$$f(x, y) = x^4 y - y - x^2 + 1$$

and classify them as local maxima, minima or saddle points.

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2. Solve the following initial-boundary value problem:

$$\begin{aligned} u_t &= u_{xx} \quad x \in (0, \pi) \quad t > 0 \\ u(0, t) &= u(\pi, t) = 0 \quad t > 0 \\ u(x, 0) &= \sin(2x) + \sin(3x) \quad x \in [0, \pi] \end{aligned}$$

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3. Find the (exact) dimensions of a cylindrical can with a lid whose volume is 1 L, and which minimises the surface area of the material used.

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4. Consider the function $f(x) = |x|$ on $-\pi \leq x \leq \pi$.

(a) Compute the Fourier series of the even periodic extension of $f(x)$ with period 2π.

(b) Use your answer from part (a) to find an exact expression for the following series

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = 1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \cdots +$$

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5. Let $\mathcal{R}_m$ be the region in the (x, y) plane bounded by the curves $y = x^m$ and $y = x$ for m an integer greater than 1.

(a) Find the centre of mass of $\mathcal{R}_m$ if the density is given as $\rho = x + y$.

(b) Using your answer from part (a) or otherwise, determine the centre of mass of $\mathcal{R}_m$ as $m \to \infty$ if the density is given as $\rho = x + y$.

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6. Write the general solution to the following ODE

$$y'' + 2y' + y = 2\cos x - 2\sin x$$

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turn to page 3

7. Let $\mathcal{S}$ be the part of the sphere $x^2 + y^2 + z^2 = 4$ between the planes $z = -1$ and $z = 1$, oriented with outward pointing normal. Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be the vector field

$$\mathbf{f}(x, y) = \left(yz + 2x, -xz + 4y^2, z^4 - 2\right).$$

(a) Draw a careful sketch of $\mathcal{S}$. Identify the boundary ∂S of S and indicate the positive orientation on ∂S induced by the outward orientation on S.

(b) Compute the curl of $\mathbf{f}$ and find the value of the surface integral

$$\iint_{\mathcal{S}} (\operatorname{curl} \mathbf{f}) \cdot \mathbf{n} \, d\mathcal{S}$$

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8. Solve the following initial boundary value problem:

$$\begin{aligned} u_{tt} &= 9u_{xx} \quad x \in (0,1) \\ u(0,t) &= u(1,t) = 0 \\ u(x,0) &= 0 \\ u_t(x,0) &= x(1-x) \qquad x \in [0,1] \end{aligned}$$

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9. Calculate the work done by the force $\mathbf{F} = x^3\mathbf{j}$ in a moving particle around the regular hexagon with vertices at $(\sqrt{3}, 1), (0, 2), (-\sqrt{3}, 1), (-\sqrt{3}, -1), (0, -2)$ and $(\sqrt{3}, -1)$ oriented anticlockwise.

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10. Find the (exact) surface area of the graph of $f(x, y) = xy$ over the region

$$\mathcal{D} = \left\{(x, y) | x^2 + y^2 \leq 1\right\}$$

(the unit disk).

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11. **Bonus:** Do not attempt this question until you have finished answering all the other questions on this exam. Tell me a joke.

This is the end of the examination paper